Tugas 2 Kriptografi

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(3)

$$W = \begin{bmatrix} AB & CD & EF & 01 \\ 23 & 45 & 67 & 89 \\ AB & CD & EF & 01 \\ 23 & 45 & 67 & 89 \end{bmatrix}$$

W[5] = W[1] ⊕ W[4]

W[4] = W[0] ⊕ g(W[3])

Rot word W[3] : 23 45 67 89

→ 45 67 89 23

Subword : 45 → 6E ⊕ 10 = 6F

⊕ Rcon[1]

67 → 85 ⊕ 00 = 85

89 → A7 ⊕ 00 = A7

23 → 26 ⊕ 00 = 26

xor with W[0] →

W[4]

AB ⊕ 6F = C4

CD ⊕ 85 = 48

EF ⊕ A7 = 48

01 ⊕ 26 = 27

} W[4]

W[5] = W[1] ⊕ W[4]

W[1] = 23 45 67 89

W[4] = C4 48 48 27 ⊕

E7 OD 2F AE

NO.

DATE

(2)

$$S = \begin{array}{c} \begin{array}{cccc} 0 & 1 & 2 & 3 \end{array} \\ \begin{bmatrix} D4 & F0 & B8 & 1E \\ BF & B4 & 41 & 27 \\ 5D & 52 & 11 & 98 \\ 30 & AE & F1 & E5 \end{bmatrix} \end{array}$$

$\downarrow$ ~~Bar~~ kolom 2

Baris 3 : [03 01 01 02]

$S'(3,2) = (03 \cdot B8) \oplus (01 \cdot 41) \oplus (01 \cdot 11) \oplus (02 \cdot F1)$

$\rightarrow 03 \cdot x = (02 \cdot x) \oplus x$

B8 = 1011 1000

$\rightarrow$ 02 x B8 = 1011 1000 $\rightarrow$ Shift left 1-bit $\rightarrow$ 0111 0000

$\rightarrow$ 0111 0000
0001 1011 $\oplus$
0110 1011 $\rightarrow$ 6B

$\rightarrow$ 03 x B8

$\rightarrow$ 6B $\oplus$ B8

$\rightarrow$ 6B = 0110 1011
B8 = 1011 1000 $\oplus$
1101 0011 $\rightarrow$ D3

$\rightarrow$ 01 x 41

$\rightarrow$ 01 . x = x

$\rightarrow$ 01 x 41 = 41

$\rightarrow$ 01 x 11 = 11

$\rightarrow$ 02 x F1

$\rightarrow$ F1 = 1111 0001

$\rightarrow$ 1111 0000 <<1 $\rightarrow$ 1110 0001

$\rightarrow$ 1110 0010
0001 1011 $\oplus$
1111 1001 $\rightarrow$ F9

D3 $\oplus$ 41 $\oplus$ 11 $\oplus$ F9

1101 0011
0100 0001
0001 0001
1111 1001 $\oplus$
0111 1010 $\rightarrow$ 7A

$S'(2,3)$ = 7A (hex)

= 0111 1010 (binary)

$x^6 + x^5 + x^4 + x^3 + x$ (polynom)

(3)

$$W = \begin{bmatrix} CA & 14 & 10 & 90 \\ FF & AC & DA & 27 \\ 83 & C1 & BF & 93 \\ 67 & 19 & E1 & 32 \end{bmatrix}$$

Rumus AES : $W[4] = W[0] \oplus g(W[3])$

where

$g(w) = \text{Subword}(\text{RotWord}(w)) \oplus \text{Rcon}[1]$

$W[3] = 67\ 19\ E1\ 92$

$\text{Rotword}(W[3]) = 19\ E1\ 92\ 67$

Subword (using S-Box)
$= 19 \rightarrow D4$
$E1 \rightarrow F8$
$32 \rightarrow 23$
$67 \rightarrow 85$

$\text{Rcon}[1] = [01\ 00\ 00\ 00]$

$g(W_3) = D4 \oplus 01 = D5$
$F8 \oplus 00 = F8$
$23\ \cancel{92} \oplus 00 = 23$
$85\ \cancel{67} \oplus 00 = 85$

$W[4] = W[0] \oplus g(W[3])$

$\rightarrow W_0 = CA\ 14\ 10\ 90$

$g(W[3]) = D5\ F8\ 23\ 85\ \oplus$

$W[4]\ 1F\ EC\ 33\ 15$